RACoN

100-Question Technical Research & EB-2 NIW Evidence Dossier

Team review draft | 27 October 2026 | Confidential - Authorized Research Team Review

Project: Resource-Aware Collaborative Network | Initiative: Metademic (described by lead as unregistered research initiative).

Repository: https://github.com/metademicteam/RACoN | Project website: https://racon.metademic.org

Evidence policy

Answers below are preliminary, not certified results. Published repository statements are distinguished from project-lead reports, proposed hypotheses and unverified experiments. No benchmark values, node-level traces, patents, immigration outcomes or third-party adoption are fabricated.

Verified here: public repository README and file index were reviewed on 27 October 2026. The actual five-PC deployment, individual tests, and full codebase were not executed or audited. The project website could not be independently retrieved during this pass.

Highest-priority discrepancy: the lead describes five-PC model partitioning, while the public README explicitly mentions distributed-runtime simulation labeling and protocol abstractions. Resolve this with synchronized per-node execution evidence before claiming demonstrated partitioned inference.

Evidence status key

PUBLIC-DOC = stated in repository documentation; LEAD-REPORTED = supplied by project lead; PROPOSED = research plan or plausible benefit; UNVERIFIED = missing independent test or document. Most answers mix statuses; evidence/action line tells the team what to prove.

Decision gates

G1: reconcile coordinator architecture and code-to-documentation claims.

G2: reproduce one real multi-node inference with shard ownership and node logs.

G3: benchmark matched single-node and 2-5 node runs, including failures and overhead.

G4: compare against prior art, test novelty and obtain independent evaluation.

G5: counsel-reviewed NIW narrative and organized dated exhibits.

Section 1: Architecture and Innovation

1. What is RACoN's architecture?

Draft answer: Local-first desktop and FastAPI coordinator, telemetry, scheduler, Ollama integration, CLI and distributed-runtime abstraction are described in README.

Verification/action: Map deployed components and commit hash.

2. Is RACoN decentralized?

Draft answer: The project lead describes per-request coordinators; published README describes a local coordinator. No fully decentralized implementation is independently established.

Verification/action: Resolve per-request vs fixed coordinator in code and diagrams.

3. How are new computers discovered?

Draft answer: Repository describes worker registration and node management; discovery across five physical machines remains unverified.

Verification/action: Capture join events and network traces.

4. How is temporary coordinator chosen?

Draft answer: User reports the requesting machine coordinates its job; current public documentation does not prove election/failover semantics.

Verification/action: Record coordinator assignment per request.

5. How are simultaneous inference requests handled?

Draft answer: Concurrency behavior cannot be established from the available public summary.

Verification/action: Run simultaneous requests with contention logs.

6. What happens on node failure?

Draft answer: Heartbeat timeout detection is documented; successful mid-inference recovery is not verified.

Verification/action: Kill a worker during inference and preserve results.

7. What differs from existing frameworks?

Draft answer: Candidate differentiator is requester-led, heterogeneous, resource-aware partitioning; novelty is not yet demonstrated.

Verification/action: Create comparison matrix against Petals, Ray, DeepSpeed, llama.cpp.

8. Does RACoN partition actual model computation?

Draft answer: The research lead reports it does; README labels distributed-runtime simulation in tests and an execution abstraction.

Verification/action: Capture model-layer or tensor ownership and actual per-node computation.

Section 2: Model Partitioning

9. What is the coordinator's exact role?

Draft answer: Proposed role: choose resources, dispatch shards, track work, aggregate outputs. Implementation scope needs code-level confirmation.

Verification/action: Provide sequence diagram and packet trace.

10. Is partitioning layer, tensor or pipeline based?

Draft answer: Not verified; terminology must follow implementation rather than intended design.

Verification/action: Document partition granularity and module references.

11. How does the partitioning algorithm work?

Draft answer: A scheduler exists, but scheduling a node is not evidence of partitioning neural-network weights or activations.

Verification/action: Provide pseudocode, source functions and trace.

12. How are CPU, RAM, GPU and VRAM considered?

Draft answer: README documents telemetry and distinction between RAM and VRAM, with unknown GPU values marked unavailable.

Verification/action: Export node telemetry at assignment time.

13. Does partition size adapt to hardware?

Draft answer: The intended resource-aware design suggests adaptation; no verified adaptive-partition test is available.

Verification/action: Compare shard allocations across heterogeneous nodes.

14. Are model weights distributed?

Draft answer: Unverified from available evidence.

Verification/action: Log weight/shard file hashes, owners and loading times.

15. Does each node execute its assigned portion?

Draft answer: User reports multiple PCs running models; independently verifiable per-node inference traces are still needed.

Verification/action: Capture operation IDs, timings and hardware counters.

16. How are intermediate tensors transferred?

Draft answer: Protocol details are not verified from README alone.

Verification/action: Document serialization, transport, tensor shapes and bytes.

17. How is execution synchronized?

Draft answer: No confirmed end-to-end synchronization protocol in reviewed evidence.

Verification/action: Capture dependency graph, barriers and timestamps.

18. How is final output assembled?

Draft answer: Proposed coordinator aggregates results; token-level output correctness needs verification.

Verification/action: Compare output with controlled baseline and fixed seed where feasible.

19. What if node memory is insufficient?

Draft answer: Scheduler hard constraints are described; behavior under a true distributed model load is unverified.

Verification/action: Test memory pressure and log fallback or rejection.

20. What proves real partitioning?

Draft answer: A passing local model call is insufficient; require shard assignments, remote compute traces and output correctness.

Verification/action: Archive correlated run ID across five nodes.

21. Which models were tested?

Draft answer: Project lead reports Gemma 3 and Dolphin model families.

Verification/action: Record exact repository/model tags and run IDs.

Section 3: Model Testing

22. Which Gemma 3 variant?

Draft answer: Not supplied; no variant can be claimed as tested.

Verification/action: Capture 1B/4B/12B/27B identifier and hash.

23. Which Dolphin variant?

Draft answer: Not supplied; Dolphin refers to multiple model releases.

Verification/action: Record exact model tag, base model and hash.

24. What are parameter count and quantization?

Draft answer: Unknown for reported tests.

Verification/action: Capture GGUF/other format, quantization, context and precision.

25. How many PCs participated per test?

Draft answer: Project lead describes a five-machine environment; participation in each single request is unverified.

Verification/action: Collect node IDs for each experiment.

26. Did all five PCs compute one inference?

Draft answer: Not independently verified; this is the highest-priority experimental question.

Verification/action: Show same run ID and distinct compute spans on each node.

27. What resource utilization occurred?

Draft answer: No raw utilization data provided.

Verification/action: Collect per-second CPU, RAM, GPU, VRAM and network traces.

28. What machine-level proof exists?

Draft answer: Public repository documentation is not a substitute for recorded five-node inference evidence.

Verification/action: Bundle synchronized logs, screenshots and profiler traces.

29. Can RACoN run a model that a single PC cannot?

Draft answer: This is a proposed benefit, not yet verified by a controlled test.

Verification/action: Document single-node OOM and successful distributed run of same model.

30. Are tests reproducible?

Draft answer: README describes an automated test suite; no independent execution of five-node model test has been done here.

Verification/action: Publish reproducible scripts and three repeat runs.

31. What is latency vs single machine?

Draft answer: Unknown; do not invent milliseconds.

Verification/action: Record end-to-end latency at matched settings.

Section 4: Performance Benchmarking

32. What is tokens per second?

Draft answer: Unknown; model and prompt must be identical.

Verification/action: Capture output token counts and elapsed generation time.

33. What is time to first token?

Draft answer: Unknown.

Verification/action: Instrument request start and first-token emission.

34. How does 2/3/4/5-node scaling behave?

Draft answer: No scaling data available.

Verification/action: Run fixed workloads with each cluster size.

35. What is network overhead?

Draft answer: Unknown; network transfer can offset compute gains.

Verification/action: Record total bytes, RTT, bandwidth and serialization time.

36. What is model loading time?

Draft answer: Unknown.

Verification/action: Separate cold start, warm start and shard-load time.

37. How does memory use change?

Draft answer: Unknown; compare peak per-node and aggregate memory.

Verification/action: Collect process-level memory peaks.

38. How much network traffic is generated?

Draft answer: Unknown.

Verification/action: Export bytes sent/received by node and job.

39. CPU-only versus GPU-assisted results?

Draft answer: Unknown.

Verification/action: Run matched CPU and GPU configurations.

40. Is distributed execution faster?

Draft answer: Not established; feasibility on memory-constrained machines may matter more than speed.

Verification/action: Report both successful capacity expansion and latency tradeoffs.

41. What baseline is appropriate?

Draft answer: Use same model, weights, quantization, prompt, token count and runtime settings; note if single-node cannot load.

Verification/action: Document hardware and fairness limits.

42. What is statistical repeatability?

Draft answer: Unknown.

Verification/action: At least three warm runs; report median, range and failures.

43. What scheduling algorithm is used?

Draft answer: README describes deterministic constraint checks, scoring objectives and optional Laya-assisted decisions.

Verification/action: Review docs/SCHEDULER.md and source implementation.

Section 5: Resource-Aware Scheduler

44. How are resources monitored?

Draft answer: README lists psutil and Windows interfaces, with non-fabrication of unknown GPU metrics.

Verification/action: Record polling method and sampling interval.

45. How often is telemetry refreshed?

Draft answer: Not established from reviewed README.

Verification/action: Inspect source/config and measure timestamps.

46. Does scheduler consider network cost?

Draft answer: High-latency benchmark scenarios are mentioned; exact scheduling formula requires inspection.

Verification/action: Trace candidate scoring with latency and bandwidth.

47. What happens when a node becomes busy?

Draft answer: Unverified for live inference.

Verification/action: Inject load and measure rescheduling or graceful failure.

48. Can workloads migrate mid-run?

Draft answer: No confirmed support.

Verification/action: Treat as future work until a passing migration test exists.

49. Are scheduling decisions reproducible?

Draft answer: Deterministic mode is described; Laya-assisted mode should be evaluated separately.

Verification/action: Save inputs, scores, decisions and random seeds.

50. How does scheduling compare with baselines?

Draft answer: No external baseline results available.

Verification/action: Compare round-robin, least-loaded and RACoN policy.

Section 6: Networking and Security

51. Does RACoN work on LAN and WAN?

Draft answer: Tailscale/Docker guidance is indexed, but working multi-site tests are unverified.

Verification/action: Run separately on LAN and VPN with measured latency.

52. What is Tailscale's role?

Draft answer: Documentation points to mesh remote access; do not infer mandatory usage.

Verification/action: Diagram transport, trust and connectivity.

53. What happens on disconnect?

Draft answer: Heartbeat timeout detection is documented, but successful job recovery is not verified.

Verification/action: Disconnect worker during model execution.

54. How is data integrity protected?

Draft answer: No verified end-to-end integrity scheme from README alone.

Verification/action: Document hashes, acknowledgments and retries.

55. Is communication encrypted?

Draft answer: Local loopback API is described with ephemeral tokens; cross-node encryption requires verification.

Verification/action: Audit TLS/Tailscale and transport config.

56. How are unauthorized nodes rejected?

Draft answer: Local API token is documented; node authentication and authorization require verification.

Verification/action: Test invalid tokens, revoked nodes and replay attempts.

57. How are prompts and model weights protected?

Draft answer: No complete threat-model conclusion can be drawn from README.

Verification/action: Document retention, transit, storage and access controls.

58. How are compromised nodes handled?

Draft answer: No independently verified malicious-node detection mechanism.

Verification/action: Perform threat model and adversarial tests.

59. How is user consent obtained for resource sharing?

Draft answer: Not established.

Verification/action: Review installer UI, consent screens and opt-out controls.

60. What are scale and security limits?

Draft answer: Unknown until stress testing and threat modeling.

Verification/action: Test node counts, bandwidth, failures and security boundaries.

Section 7: Software and GitHub Evidence

61. What code components are complete?

Draft answer: Repository visibly contains apps/desktop, backend, CLI, installer, docs and tests.

Verification/action: Create source inventory by commit hash.

62. What is simulated versus real?

Draft answer: README explicitly refers to distributed runtime simulation labeling; exact boundaries need audit.

Verification/action: Mark simulated paths in architecture and claims ledger.

63. Does README match code?

Draft answer: Not established by a full code audit.

Verification/action: Review each capability against executable tests.

64. What do automated tests cover?

Draft answer: README lists 80 unit/integration tests spanning API, telemetry, scheduler, simulation and state transitions; not independently run here.

Verification/action: Execute pytest in clean environment and save CI output.

65. Are end-to-end multi-PC tests present?

Draft answer: Not established by README.

Verification/action: Add five-node end-to-end model-partition test.

66. Does installer work on clean Windows?

Draft answer: Windows launcher/installer files exist; clean-install success is unverified.

Verification/action: Run signed/unsigned install tests on fresh Windows VM.

67. How is release history preserved?

Draft answer: Repository contains changelog and git history; current release provenance needs capture.

Verification/action: Archive tag, commit, build hashes and changelog.

68. Who contributed what?

Draft answer: Not verified; repository ownership alone does not establish each researcher's contribution.

Verification/action: Create dated contribution ledger and signed attestations.

69. What is license and IP ownership?

Draft answer: LICENSE file exists, but legal ownership and contributor assignments need review.

Verification/action: Inspect license and contributor agreements with counsel.

70. Can outsiders reproduce results?

Draft answer: Documentation and CLI exist, but reproducible five-node results not yet established.

Verification/action: Provide setup, environment lockfile, scripts and anonymized data.

Section 8: Scientific Novelty and Publications

71. What research problem is addressed?

Draft answer: Potential problem: limited access to large AI inference on heterogeneous consumer hardware.

Verification/action: Define measurable target population and technical constraints.

72. What gap in prior work exists?

Draft answer: A gap is hypothesized, not yet proven.

Verification/action: Systematic literature review with dated comparison table.

73. How does RACoN compare with Petals, Ray, DeepSpeed and llama.cpp?

Draft answer: A direct benchmark is not available; these tools differ in orchestration and distributed inference capabilities.

Verification/action: Compare equivalent workloads and capabilities fairly.

74. Which components are novel?

Draft answer: Novelty is unproven; possible claims include requester-led coordination and adaptive partitioning.

Verification/action: Identify algorithmic differences and supporting ablations.

75. What is borrowed versus original?

Draft answer: Not documented in current evidence.

Verification/action: Map dependencies, prior art and original commits.

76. What is the main hypothesis?

Draft answer: Proposed: capacity-aware partitioning enables correct inference of models that exceed one node's memory, within acceptable overhead.

Verification/action: Pre-register hypothesis and success thresholds.

77. How will the hypothesis be tested?

Draft answer: Controlled same-model single-node and multi-node trials, with correctness, memory and latency outcomes.

Verification/action: Write test protocol before collecting final data.

78. Which public datasets/workloads?

Draft answer: No standard workload chosen.

Verification/action: Choose reproducible prompt set and fixed token limits; document licensing.

79. What are negative results?

Draft answer: Not yet provided; absence of data is not evidence of no failures.

Verification/action: Preserve failed runs, overhead and limitations.

80. Is research publication-ready?

Draft answer: Not yet: architecture reconciliation, independent replication and benchmarks are outstanding.

Verification/action: Draft paper after technical validation.

Section 9: U.S. National Importance

81. Which U.S. problem could RACoN address?

Draft answer: Potential access barriers to compute-intensive AI research and deployment.

Verification/action: Use documented U.S. needs, not generic slogans.

82. How could universities and SMEs benefit?

Draft answer: Potentially use existing heterogeneous hardware; benefit has not been measured.

Verification/action: Pilot with an independent U.S. institution.

83. Could it reduce infrastructure cost?

Draft answer: Plausible hypothesis, not demonstrated.

Verification/action: Compare hardware, power, network, labor and maintenance costs.

84. How will cost savings be measured?

Draft answer: Total cost of ownership and cost per successful inference are appropriate metrics.

Verification/action: Record electricity, hardware amortization, support and throughput.

85. Could impact extend beyond one company?

Draft answer: Open tooling and multi-institution adoption could support broader impact; current adoption unverified.

Verification/action: Collect external pilots, users and citations.

86. Which U.S. sectors are relevant?

Draft answer: Potentially education, research, small businesses and edge computing.

Verification/action: Prioritize sectors with real pilot interest.

87. Is there independent interest?

Draft answer: No verified external letters or adoption evidence supplied.

Verification/action: Request specific evaluation letters and pilot results.

88. Can pilot deployments be arranged?

Draft answer: Potentially, subject to permission and F-1 work authorization review.

Verification/action: Define pilot scope, host and compliance checks.

89. What is 1-3 year roadmap?

Draft answer: Propose validation, open reproducible release, independent pilots, scaling and publications.

Verification/action: Tie milestones to budget and responsible owners.

90. What resources and partners are needed?

Draft answer: Compute nodes, benchmarking expertise, security review, institutional pilots and funding.

Verification/action: Create resource plan and support letters.

Section 10: Researcher and NIW Evidence

91. What is lead researcher's contribution?

Draft answer: Project lead reports conceptual leadership and architecture; documentary proof is needed.

Verification/action: Collect dated design notes, commits, meeting records.

92. Which papers directly support RACoN?

Draft answer: Researcher reports publications; specific relevant papers not yet cross-mapped.

Verification/action: Create publication-to-endeavor matrix with DOI.

93. Which citations are relevant?

Draft answer: Reported total citations do not show RACoN-specific influence.

Verification/action: Use dated Google Scholar export and relevant citing works.

94. What independent recognition exists?

Draft answer: Professional roles are reported; their direct relevance requires documentation.

Verification/action: Collect invitations, review records and external references.

95. How do memberships support this endeavor?

Draft answer: IEEE and other memberships may provide context but are not automatic proof of NIW eligibility.

Verification/action: Document selection criteria and research relevance.

96. Is there a relevant patent?

Draft answer: Researcher reports inventions; RACoN-specific filing and ownership not verified.

Verification/action: Gather filing numbers, dates, claims and ownership.

97. Can experts assess significance?

Draft answer: Independent experts may evaluate the specific technical contribution.

Verification/action: Seek evidence-based letters, not generic praise.

98. How does experience support execution?

Draft answer: Reported background spans analytics, finance, telecom and technology research.

Verification/action: Link verified accomplishments to concrete RACoN milestones.

99. What proves future execution ability?

Draft answer: Prototype, team plan and past results may help; budget and collaborations need substantiation.

Verification/action: Prepare 3-year milestones, resources and letters.

100. Why waive job offer and labor certification?

Draft answer: A potential argument is that independent cross-institutional research could benefit the U.S.; a waiver is not automatic.

Verification/action: Counsel should assess all Dhanasar prongs and immigration facts.

12-week execution plan

Weeks 1-2: Code-to-claim audit; architecture diagram; version/commit freeze; owner: technical lead.

Weeks 3-4: Five-node experiment setup; model hashes; run IDs; per-node tracing; owner: ML engineering.

Weeks 5-6: Same-model benchmarks (1, 2, 3, 4, 5 nodes), three or more repeats, failed runs; owner: QA/performance.

Weeks 7-8: Threat model, disconnect tests, security review and reproducibility; owner: security lead.

Weeks 9-10: Prior-art comparison, cost model, independent pilots and impact evidence; owner: research lead.

Weeks 11-12: Exhibits, researcher contribution records, proposed endeavor and counsel review; owner: petition team.

Minimum benchmark record

experiment_id | timestamp/timezone | git_commit | model_id/hash | quantization | prompt_set | context | output_tokens | seed/temperature | node_specs | network_RTT/bandwidth | shard_map | per_node_compute_trace | latency_ms | ttft_ms | tokens_per_sec | peak_RAM/VRAM | bytes_transferred | outcome | error_logs | reviewer

NIW framing (draft, not a legal conclusion)

Proposed endeavor: advance resource-aware distributed AI inference across heterogeneous hardware to investigate whether compute-constrained U.S. researchers and organizations can access larger models using existing machines. Demonstrate substantial merit and national importance through reproducible technical results and evidence of broader prospective impact; show the petitioner is well positioned through verifiable contributions, publications and collaboration; separately analyze the overall benefit of waiving job offer and labor certification under Matter of Dhanasar.

Compliance: Metademic should not be described as a registered nonprofit absent registration. F-1 work authorization and permitted research activities should be reviewed with DSO and qualified immigration counsel before external commercial work or pilots. Do not make unsupported adoption, cost-saving or patent claims.

Public sources and provenance

GitHub repository README and index: https://github.com/metademicteam/RACoN (accessed 27 Oct 2026).

GitHub distributed-runtime documentation listed in README: https://github.com/metademicteam/RACoN/blob/main/docs/DISTRIBUTED_RUNTIME.md (referenced in prior discussion; exact contents require version-controlled review).

Team-reported statements: five heterogeneous PCs; Gemma 3 and Dolphin tests; requester-as-coordinator design; model-partitioning objective. These remain lead-reported pending original logs.